

Machine Learning Notes: Linear Regression, Random Forest & XGBoost

Machine Learning Notes

Linear Regression, Random Forest & XGBoost

1. Linear Regression

Linear Regression is a supervised machine learning algorithm used for predicting continuous numerical values.

Linear Regression Equation:

$$y = mx + b$$

Applications:

- House price prediction
- Salary prediction
- Sales forecasting
- Temperature prediction

Advantages:

- Simple and easy to understand
- Fast training
- Good for linear relationships

Disadvantages:

- Poor performance on complex datasets
- Sensitive to outliers
- Assumes linear relationship

2. Random Forest Regression

Random Forest is an ensemble learning algorithm based on multiple decision trees.

It combines predictions from many trees to improve accuracy and reduce overfitting.

How Random Forest Works:

- 1. Multiple random samples are created**
- 2. Separate decision trees are trained**
- 3. Each tree makes predictions**
- 4. Final output is the average prediction**

Important Parameter:

`n_estimators = 200`

Advantages:

- High accuracy
- Reduces overfitting
- Handles large datasets well
- No scaling required

Disadvantages:

- Higher memory usage
- Slower than simple models

3. XGBoost

XGBoost stands for Extreme Gradient Boosting.

It is an advanced boosting algorithm used for regression and classification.

How XGBoost Works:

- 1. First tree makes predictions**
- 2. Errors are calculated**
- 3. Next tree corrects previous errors**
- 4. Process repeats multiple times**

Important Parameters:

- `n_estimators`
- `learning_rate`
- `max_depth`

Advantages:

- Very high accuracy
- Handles complex data well
- Built-in regularization

Disadvantages:

- More complex
- Requires tuning
- Higher computational cost

4. Evaluation Metrics

MAE (Mean Absolute Error)

Measures average prediction error.

MSE (Mean Squared Error)

Measures squared prediction error.

R^2 Score

Measures how well the model fits the data.

Closer to 1 means better performance.

5. Difference Between Algorithms

Linear Regression:

- Simple algorithm
- Uses straight-line relationship
- Faster training

Random Forest:

- Uses multiple trees
- High accuracy
- Reduces overfitting

XGBoost:

- Sequential learning
- Corrects previous errors
- Very high accuracy

6. California Housing Dataset

The California Housing dataset is used for predicting house prices.

Features:

- Median Income
- House Age
- Average Rooms
- Population
- Latitude
- Longitude

Target:

- Median House Value

7. Conclusion

Linear Regression is simple and suitable for basic prediction tasks.

Random Forest improves accuracy using multiple decision trees.

XGBoost further improves performance by correcting previous errors sequentially.

These algorithms are widely used in:

- Price prediction
- Forecasting
- Financial analysis
- Data science projects